



Professional and Continuing Education

Customer Segmentation with Clustering

An unsupervised machine learning analysis of 68,300 customers across 47 countries, identifying five distinct behavioural segments to support a targeted marketing strategy.

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Table of Contents

Introduction.....	2
Dataset.....	2
Methods.....	2
Exploratory Data Analysis (EDA).....	3
Feature Engineering & Scaling.....	4
Clustering.....	4
Dimensionality Reduction.....	4
Results & Discussion.....	5
EDA Results.....	5
K-means Clustering Results.....	6
PCA & t-SNE Results.....	7
Conclusion & Recommendations.....	8

Introduction

Understanding customers is one of the most valuable things a business can do. This project tackles customer segmentation for a global e-commerce company, using machine learning to group customers based on their behaviour and demographics. Using clustering techniques, the project will create customer segmentation groups for business analysis. The goal is to give the business a clearer picture of who their customers are, so they can market to them more effectively, retain the valuable ones, and re-engage those who have gone quiet.

Dataset

The dataset was provided by SAS (2024) and contains 951,669 rows and 20 columns. Each row represents a single product ordered by a customer, which meant the data needed to be aggregated before any analysis could be done. The orders span from 1 January 2012 to 30 December 2016, covering customers from 47 countries across five continents.

Methods

This project followed a structured analytical pipeline, progressing from initial data exploration and cleaning through to feature engineering, clustering, and dimensionality reduction. The goal was to identify distinct customer segments from behavioural and demographic data, enabling targeted marketing recommendations. The full pipeline is illustrated in Figure 1, and key methods and outputs are summarised in Table 1.

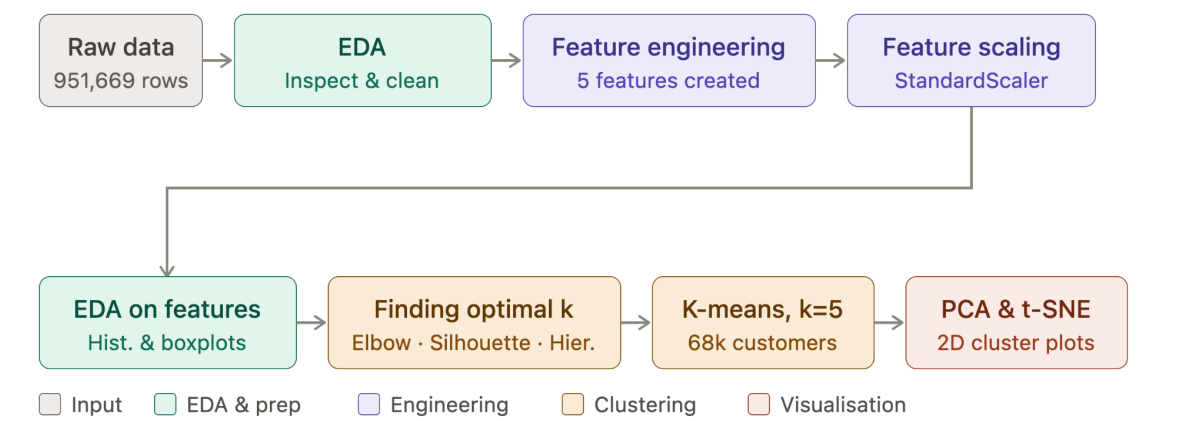


Figure 1: Analysis pipeline from raw data to customer segments.

Table1. Summary of analytical methods and key outputs.

Step	Method	Detail
EDA	info(), isnull(), drop_duplicates()	Shape, dtypes, missing values, duplicates identified
Data Cleaning	Currency conversion, datetime parsing	21 duplicates removed, \$ signs stripped
Feature Engineering	groupby().agg()	5 features, 68,300 unique customers
Feature Scaling	StandardScaler	Mean=0, std=1
Finding Optimal k	Elbow, Silhouette, Hierarchical	All three methods → k=5
K-means Clustering	KMeans(k=5)	Applied to the full dataset
Dimensionality Reduction	PCA, t-SNE	2D cluster visualisation

Exploratory Data Analysis (EDA)

Initial EDA was performed on the raw dataset to assess data quality and understand its structure. This involved inspecting and cleaning the dataset's shape, data types, and non-null counts. Following feature engineering, a second EDA was performed on the five engineered features. Descriptive statistics were used to assess central tendency and skewness, histograms with KDE curves were plotted to visualise the distribution of each feature, and boxplots were used to identify outliers across customer segments.

Feature Engineering & Scaling

From the raw dataset, five customer-level features were created: Frequency, Recency, CLV, Average Unit Cost, and Age. Table 2 summarises the source column and aggregation operation for each feature. As these features operate on significantly different scales, StandardScaler was applied to all five features, transforming each to have a mean of 0 and a standard deviation of 1 to ensure equal contribution to the clustering process.

Table 2. Source and operation of the engineered features.

Feature	Source	Operation
Frequency	Order ID	count
Recency	Order_Date	days since last order (ref: 30 Dec 2016)
CLV	Total Revenue	sum
Avg_Unit_Cost	Unit Cost	mean
Age	Customer_BirthDate	First occurrence → then calculate age

Clustering

Due to computational demands, the scaled data was reduced to a random sample of 30,000 customers for k-selection methods. Three approaches were applied: the Elbow method (WCSS), Silhouette score, and Hierarchical clustering with a dendrogram. All three converged on k=5. K-means was then applied to the full dataset of 68,300 customers, with cluster assignments recorded per Customer ID. Boxplots were generated for each feature split by cluster to enable visual comparison across segments.

Dimensionality Reduction

To visualise the five-dimensional customer segments in 2D, two dimensionality reduction techniques were applied to the full-scale dataset. Principal Component Analysis (PCA) was used as a linear method to reduce the data to two principal components, preserving global structure. t-SNE was also applied as a non-linear alternative, with perplexity set to 30, which better preserves local neighbourhood structure and cluster boundaries. The resulting 2D scatter plots were coloured by cluster label to evaluate the separation of customer segments.

Results & Discussion

EDA Results

EDA on the five engineered features revealed clear distributional patterns. Frequency, CLV, Average Unit Cost, and Recency are all right-skewed, with means notably higher than medians (eg, CLV mean 1,950 vs median 1,497), indicating that a small number of high-value customers drive disproportionately high values. Customer Age shows a near-identical mean (43.13) and median (43.0), confirming a uniform distribution across ages 20–75. Boxplots identified outliers in Frequency, CLV, Average Unit Cost, and Recency. As these represent genuine customer behaviour rather than data errors, they were retained to preserve the integrity of the segmentation. Distributional patterns are illustrated in Figure 2.

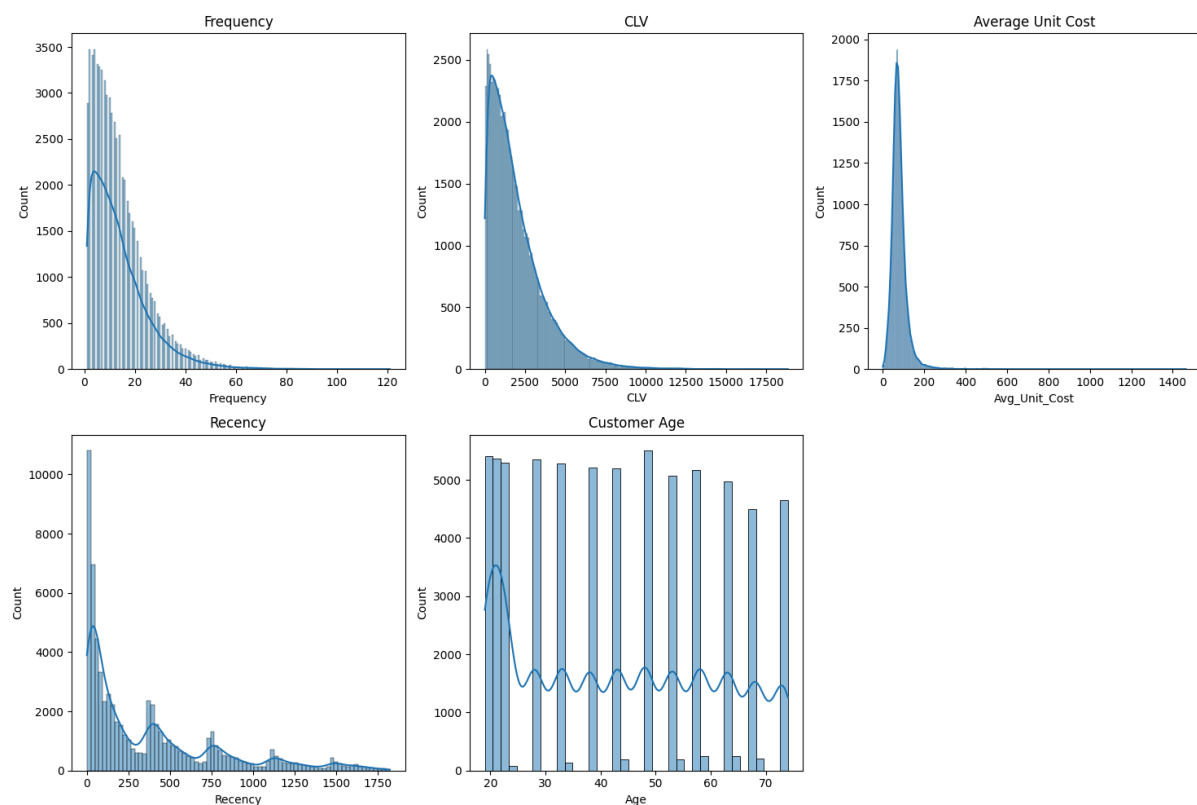


Figure 2: Distribution of the five engineered features.

K-means Clustering Results

The optimal number of clusters ($k=5$) was confirmed by three independent methods: the Elbow method, the Silhouette score (peak at 0.267 at $k=5$; Figure 3), and the Hierarchical clustering dendrogram, with all three methods converging on the same value. K-means was applied to the full dataset of 68,300 customers, producing five segments of varying sizes (Figure 4).

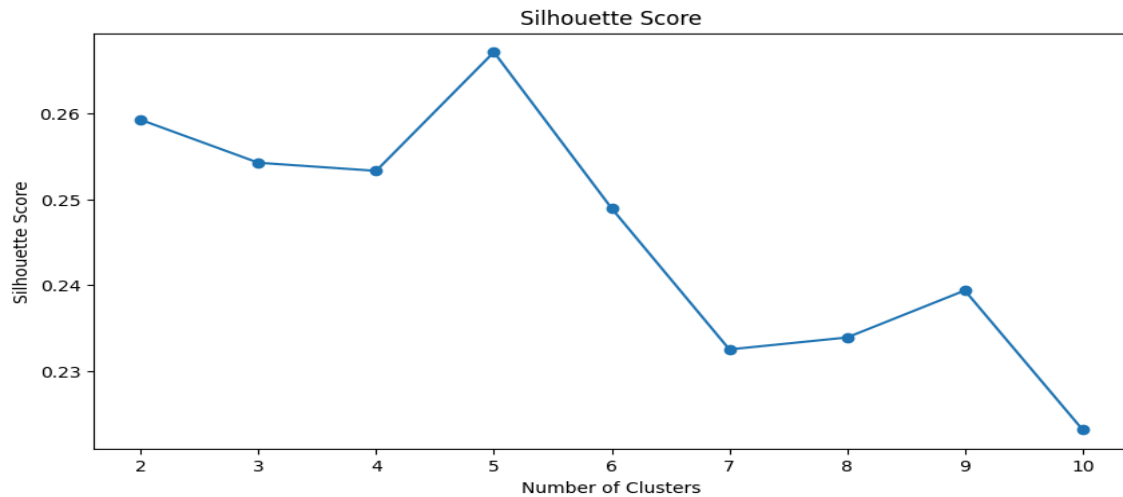


Figure 3: Silhouette scores for k=2 to k=10.

Analysis of cluster boxplots revealed five distinct customer profiles. Cluster 0 (Older Budget Buyers) showed low CLV and moderate Frequency, with a median age of ~57 years. Cluster 1 (Younger Budget Buyers) had similar spending behaviour but a median age of ~28 years. Cluster 2 (Churned/Inactive Customers) had the highest Recency (~1,100 days), indicating long-term disengagement. Cluster 3 (High-Value Loyal Customers) recorded the highest Frequency and CLV, confirming them as the most valuable segment. Cluster 4 (Premium Occasional Buyers) showed the highest Average Unit Cost but low Frequency, representing infrequent high-spending customers. The cluster profiles are further illustrated in Figure 4 below.

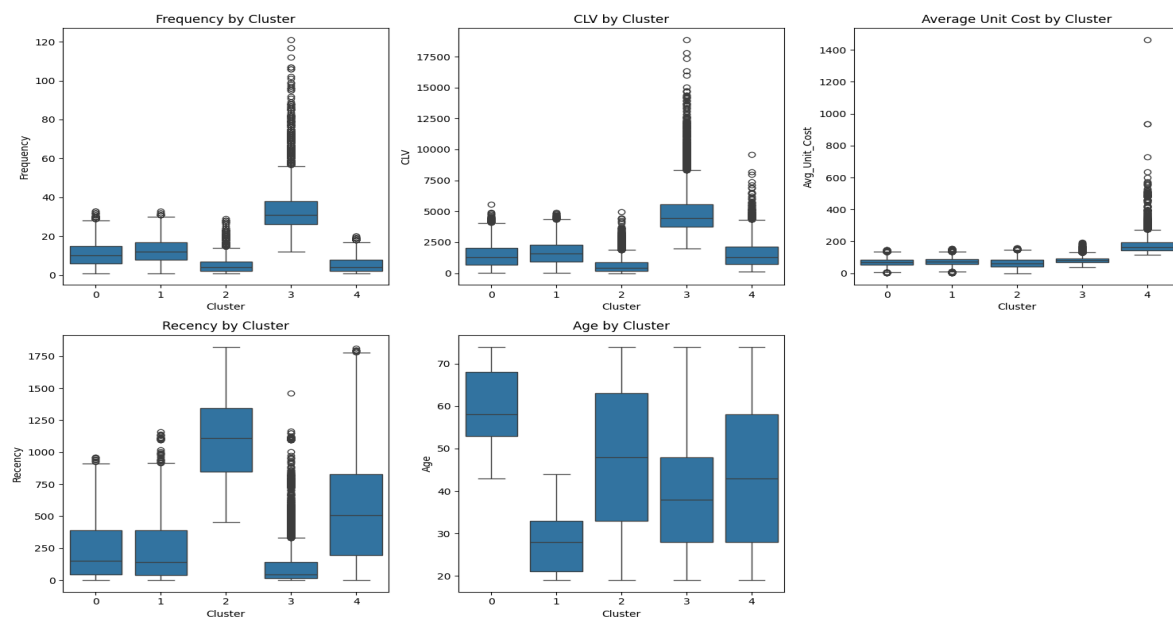


Figure 4: Cluster boxplots showing feature distributions across five customer segments.

PCA & t-SNE Results

PCA reduced the five features to two components, explaining 65.3% of the total variance (Figure 5). The resulting scatter plot showed partial cluster separation with some overlap, expected given the 34.7% information loss in compression. Cluster 4 was visibly separated on the y-axis, consistent with its distinctively high Average Unit Cost.

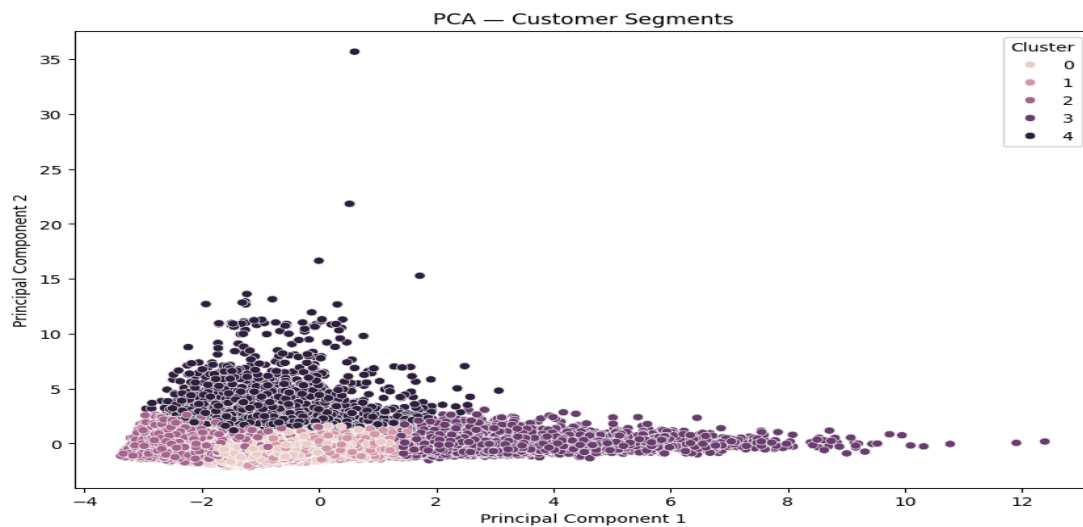


Figure 5: PCA 2D visualisation of customer segments.

t-SNE produced noticeably clearer cluster separation (Figure 6), with each segment occupying more distinct regions of the plot. However, t-SNE took approximately 26 minutes to run compared to 1-2 minutes for PCA, and its axes carry no interpretable meaning.

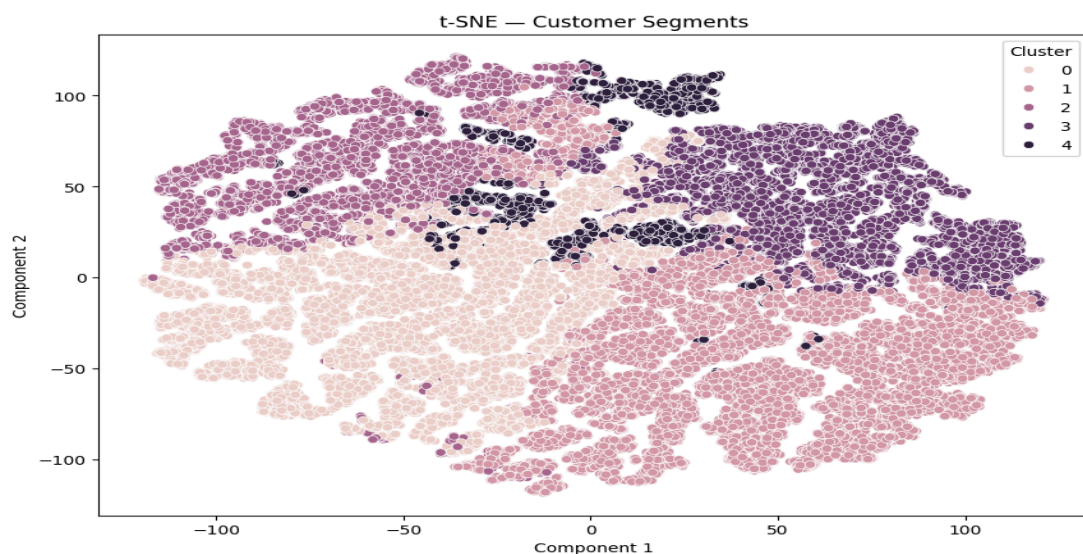


Figure 6: t-SNE 2D visualisation of customer segments.

Overall, t-SNE provided superior visual clarity for cluster structure, while PCA offered a faster and more quantifiable alternative.

Conclusion & Recommendations

This project successfully segmented 68,300 customers into five distinct groups using K-means clustering, validated by three independent methods. The analysis revealed meaningful behavioural and demographic differences between segments, providing a foundation for targeted marketing strategies.

Based on the cluster profiles, the following recommendations are proposed in the table below.

Table 3: Marketing recommendations by customer segment.

Cluster	Segment	Recommendation
0	Older Budget Buyers	Loyalty rewards and value-focused promotions/bundle deals and discounts to increase CLV
1	Younger Budget Buyers	Digital and social media campaigns with trend-led, affordable products and influencer marketing
2	Churned/Inactive Customers	Win-back campaigns with personalised re-engagement offers and time-sensitive discount codes
3	High-Value Loyal Customers	VIP programmes, early product access, and personalised communication to protect retention
4	Premium Occasional Buyers	Exclusive high-end product launches and luxury promotions to increase purchase frequency

A key limitation of this analysis is that only five features were used for segmentation. Incorporating additional variables such as geographic location, product category

preferences, or seasonal purchasing patterns could further refine the customer profiles and improve targeting precision. Future work could also incorporate more recent data to better reflect current customer behaviour.